

الجامعة الافتراضية السورية

Syrian Virtual University

بكالوريوس تكنولوجيا المعلومات

Bachelor of Information Technology

مقرر:

ITI 410 (Data Security)

الفصل : F 11

Chapter : F 11

الصف : C 1 - C3

Class : C 1 - C3

اشراف : د. حسام عبيد

Dr. Houssam Abed



...الموضوع...

The assignment aims at enforcing the cryptography techniques. It helps the student in acquiring the necessary knowledge in understanding cryptography applications and in securing data using encryption techniques.

Objectives can be summarized as:

- Understanding cryptography applications
- Encrypting and decrypting using DES and RSA
- Hashing using SHA1

Feras_38850

Moayad_30605

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Dima_28679

Mhmad_hani_36220

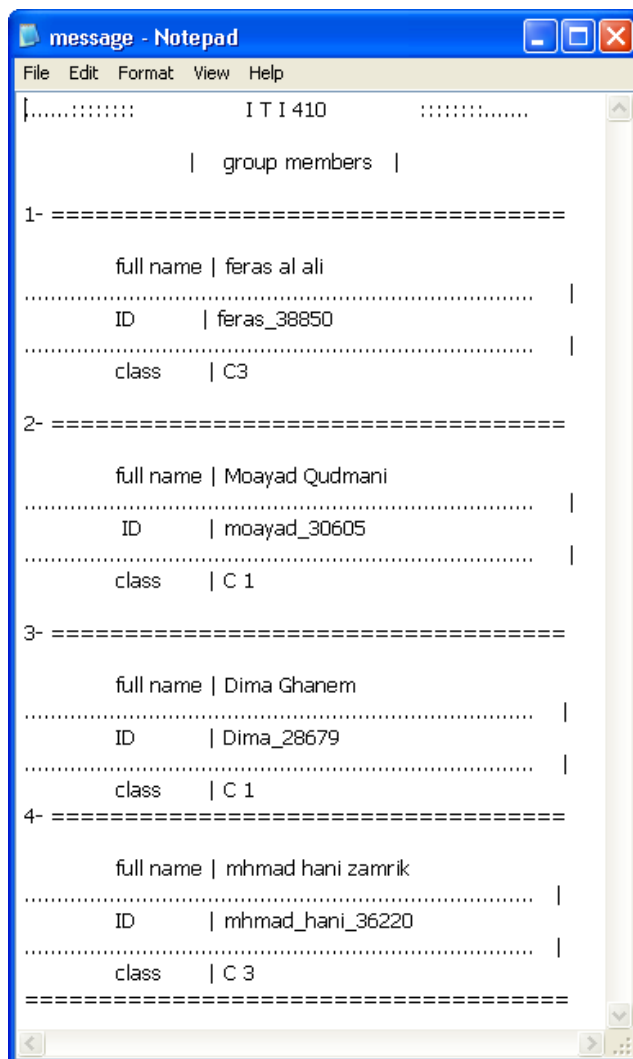
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- We've created folder Contains all the requirements of files for this assignment, named:
\\SVU_BIT_F11_ITI410_Assignment_C1_C3_Feras_38850_Moayad_30605_Dima_28679_Mhmad_hani_36220

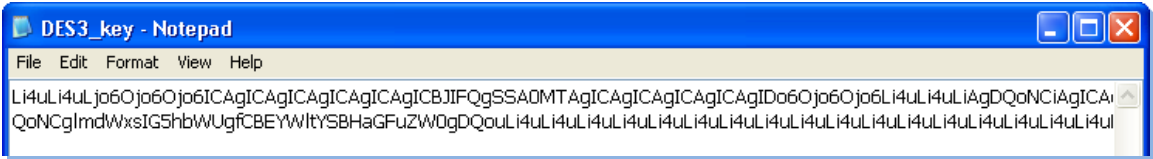
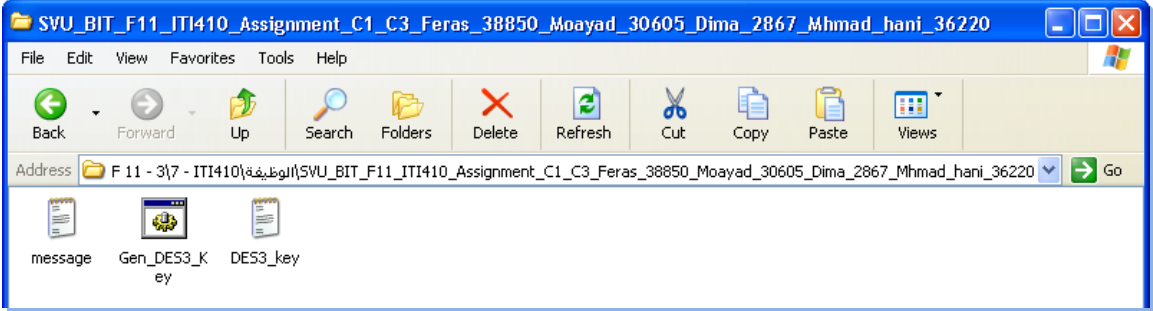
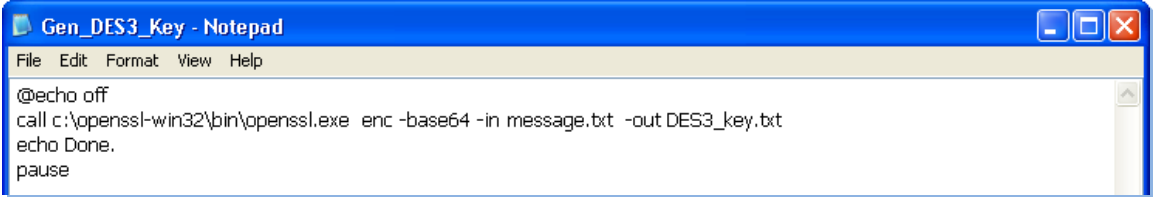


- 1- Create a text file called "message.txt" containing the full name, ID and class of each one of the group members.



2- Write a command line in a batch file called "Gen_DES3_Key.bat" to generate a DES3 key coded in base64. Run the command and save the key in a file called "DES3_key.txt".

```
@echo off
call c:\openssl-win32\bin\openssl.exe enc -base64 -in message.txt -out
DES3_key.txt
echo Done.
pause
```

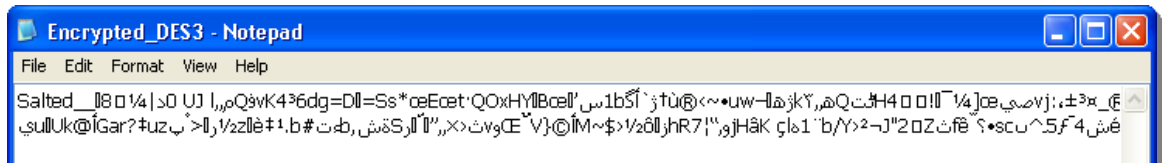
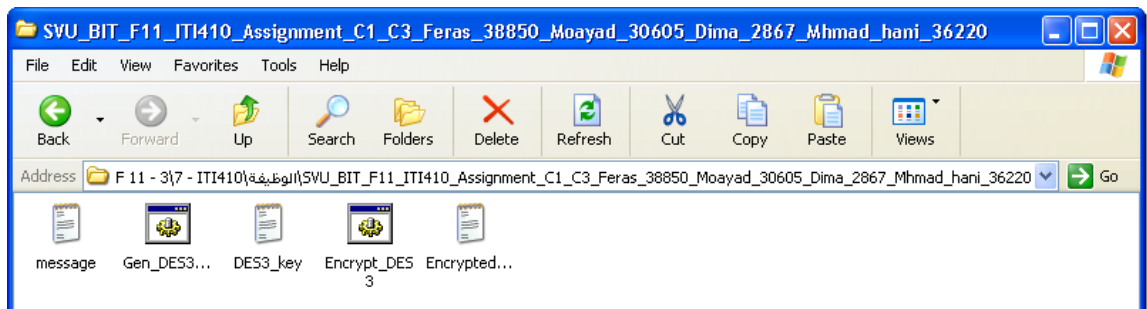
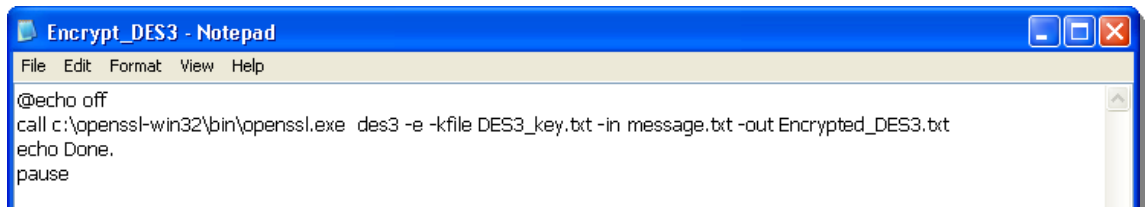


code	Details
@echo off	command will either turn ON, or OFF, you put in a batch file from showing itself
call Path file	call file or program using full path
enc	Encryption command , Standard commands
-base64	Encryption Option
-in file	options for getting file of available from path or name
-out file	option or your shell to redirect it to a file
echo Done.	End echo
pause	Following the initial stop work when the user clicks on any button

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- 3- Write a command line in a batch file called "Encrypt_DES3.bat" to encrypt the file "message.txt" using the DES3 algorithm and the key stored in "DES3_key.txt". Run the command and save the result in a file called "Encrypted_DES3.txt".

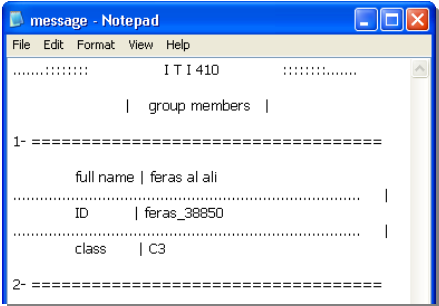
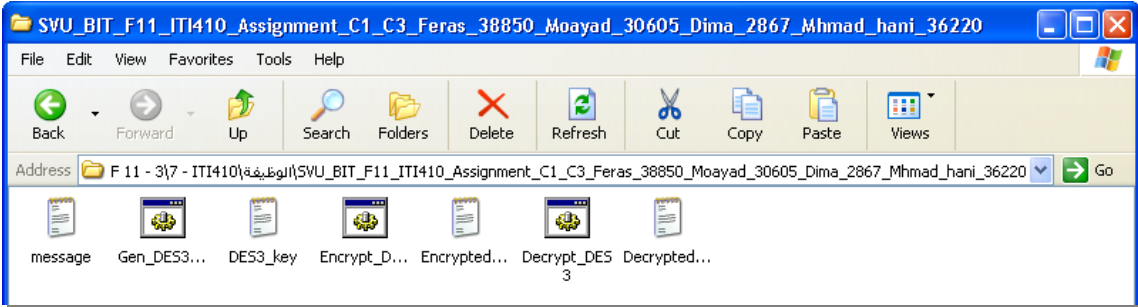
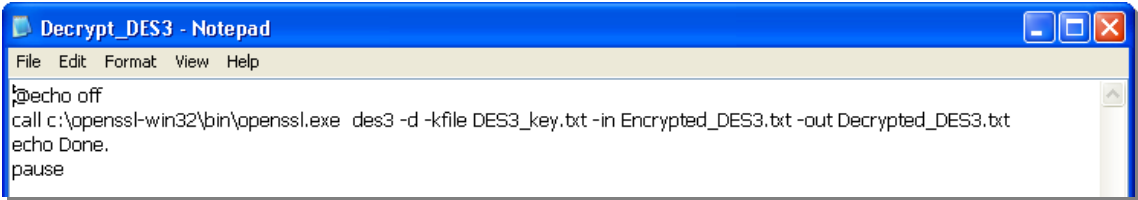
```
@echo off
call c:\openssl-win32\bin\openssl.exe des3 -e -kfile DES3_key.txt -in
message.txt -out Encrypted_DES3.txt
echo Done.
pause
```



code	Details
@echo off	command will either turn ON, or OFF, you put in a batch file from showing itself
call Path file	call file or program using full path
des3	The functionality includes symmetric encryption,
-e	use engine e, possibly a hardware device
-kfile file	Single-key DES is insecure due to its short key size
-in file	options for getting file of available from path or name
-out file	option or your shell to redirect it to a file
echo Done.	End echo
pause	Following the initial stop work when the user clicks on any button

4- Write a command line in a batch file called "Decrypt_DES3.bat" to decrypt the file "Encrypted_DES3.txt" using the DES3 algorithm and the key stored in "DES3_key.txt". Run the command and save the result in a file called "Decrypted_DES3.txt".

```
@echo off
call c:\openssl-win32\bin\openssl.exe des3 -d -kfile DES3_key.txt -in Encrypted_DES3.txt -out Decrypted_DES3.txt
echo Done.
pause
```



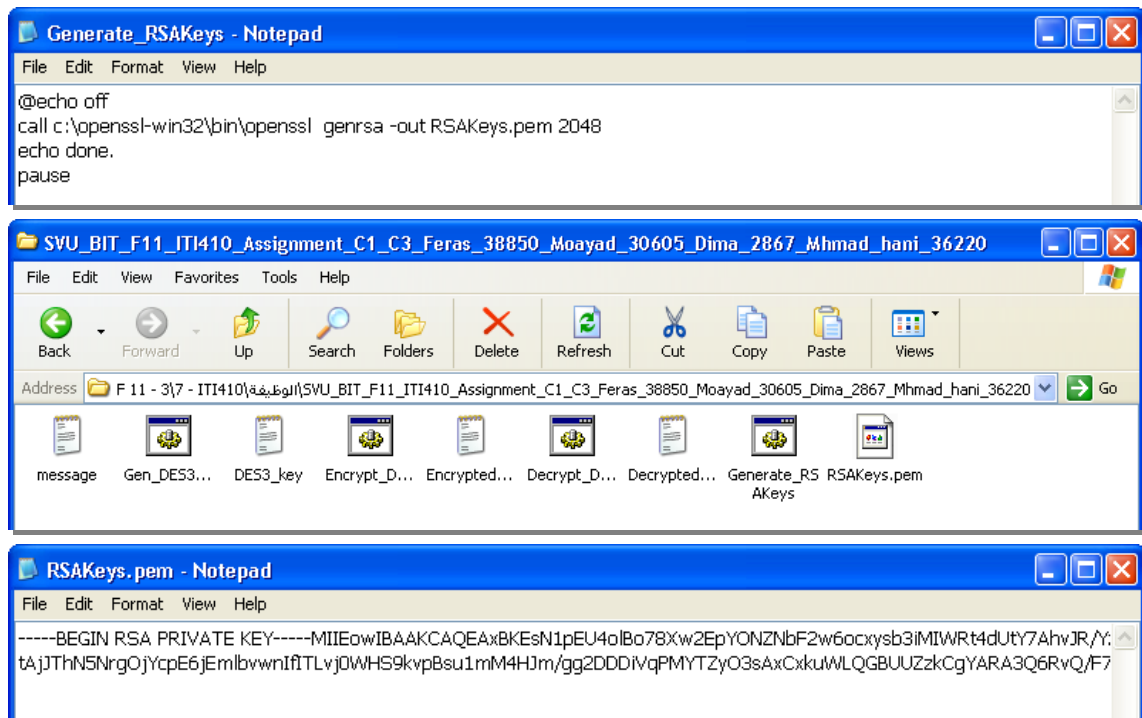
code	Details
@echo off	command will either turn ON, or OFF, you put in a batch file from showing itself
call Path file	call file or program using full path
des3	The functionality includes symmetric encryption,
-d	to output debug info
-kfile	Single-key DES is insecure due to its short key size
-in file	options for getting file of available from path or name
-out file	option or your shell to redirect it to a file
echo Done.	End echo
pause	Following the initial stop work when the user clicks on any button

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5- Write a command line in a batch file called "Generate_RSAKeys.bat" to generate RSA private/public keys of 2048-bits length. Run the command and save the RSA keys in a file called "RSAKeys.pem".

```
@echo off
call c:\openssl-win32\bin\openssl genrsa -out RSAKeys.pem 2048
echo done.
Pause
```

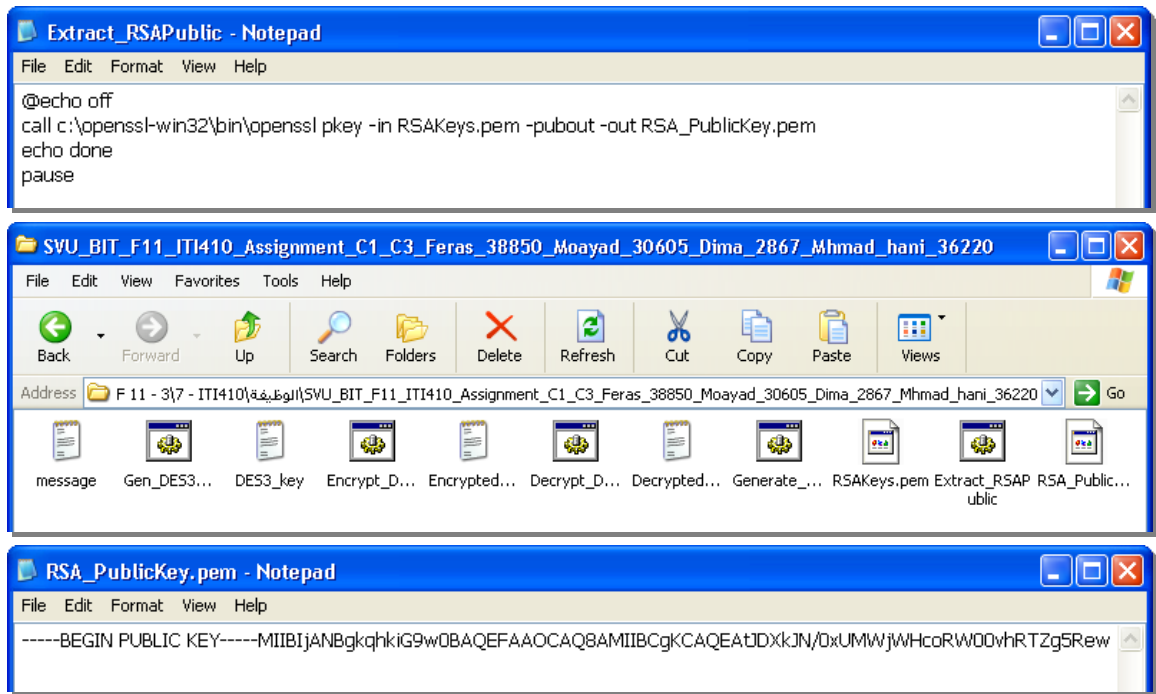


code	Details
@echo off	command will either turn ON, or OFF, you put in a batch file from showing itself
call Path file	call file or program using full path
genrsa	Standard commands , generate an RSA key
-out file	option or your shell to redirect it to a file
echo Done.	End echo
pause	Following the initial stop work when the user clicks on any button

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6- Write a command line in a batch file called "Extract_RSAPublic.bat" to extract the RSA public key from the RSA keys file "RSAKeys.pem". Run the command and save the public key in a file called "RSA_PublicKey.pem".

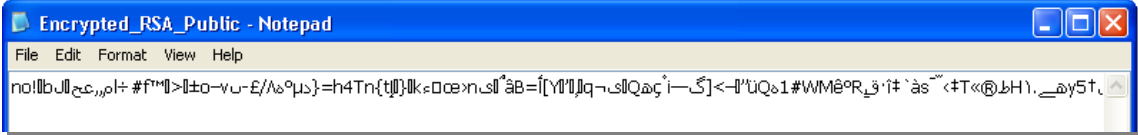
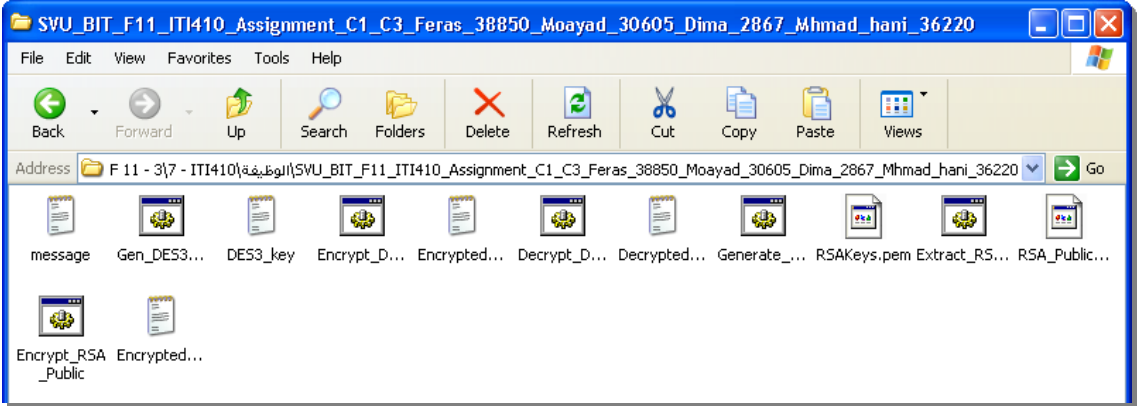
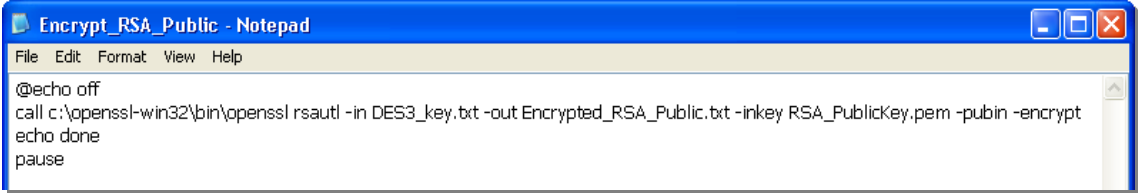
```
@echo off
call c:\openssl-win32\bin\openssl pkey -in RSAKeys.pem -pubout -out
RSA_PublicKey.pem
echo done
pause
```



code	Details
@echo off	command will either turn ON, or OFF, you put in a batch file from showing itself
call Path file	call file or program using full path
pkey	public or private key processing tool
-pubout	default a private key is output: with this option a public key will be output instead.
-in file	options for getting file of available from path or name
-out file	option or your shell to redirect it to a file
echo Done.	End echo
pause	Following the initial stop work when the user clicks on any button

7- Write a command line in a batch file called "Encrypt_RSA_Public.bat" to encrypt the file "DES3_key.txt" using the RSA public key. Run the command and save the result in a file called "Encrypted_RSA_Public.txt".

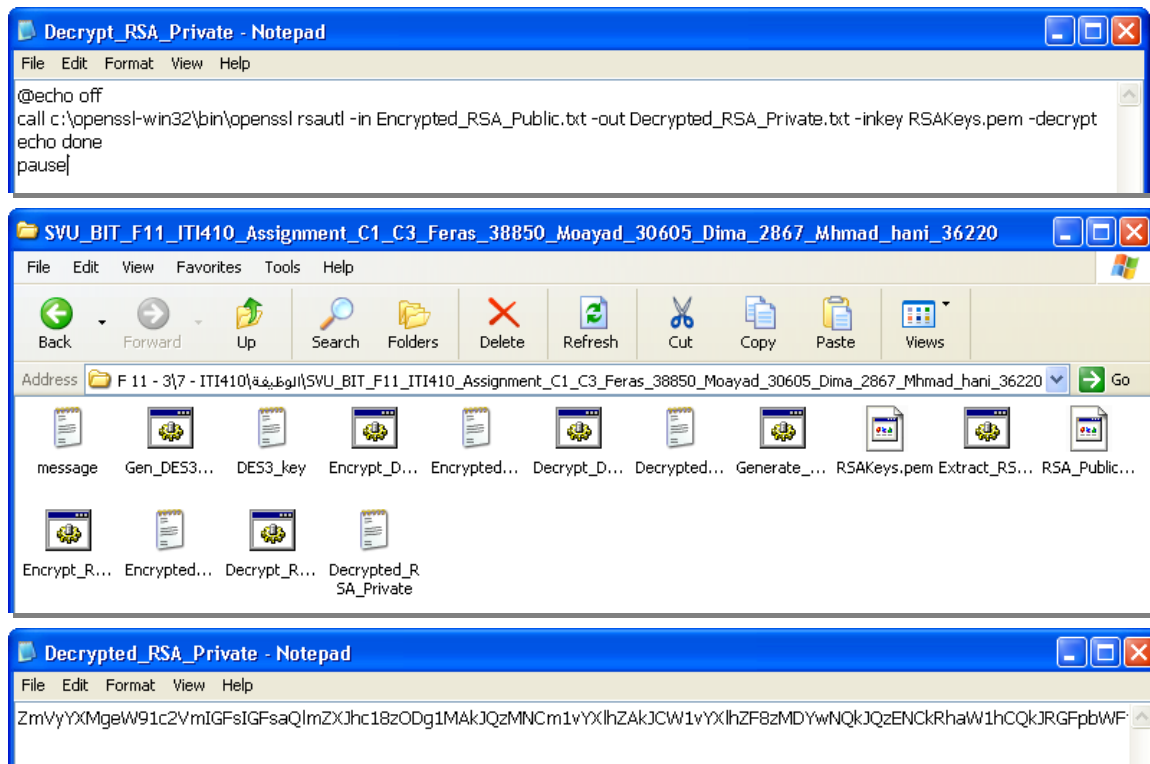
```
@echo off
call c:\openssl-win32\bin\openssl rsautl -in DES3_key.txt -out
Encrypted_RSA_Public.txt -inkey RSA_PublicKey.pem -pubin -encrypt
echo done
pause
```



code	Details
@echo off	command will either turn ON, or OFF, you put in a batch file from showing itself
call Path file	call file or program using full path
rsautl	RSA utility
-in file	options for getting file of available from path or name
-out file	option or your shell to redirect it to a file
-inkey file	the input key file, by default it should be an RSA private key
-pubin	the input file is an RSA public key
-encrypt	encrypt the input data using an RSA public key
echo Done.	End echo
pause	Following the initial stop work when the user clicks on any button

- 8- Write a command line in a batch file called "Decrypt_RSA_Private.bat" to decrypt the file "Encrypted_RSA_Public.txt" using the RSA private key. Run the command and save the result in a file called "Decrypted_RSA_Private.txt".

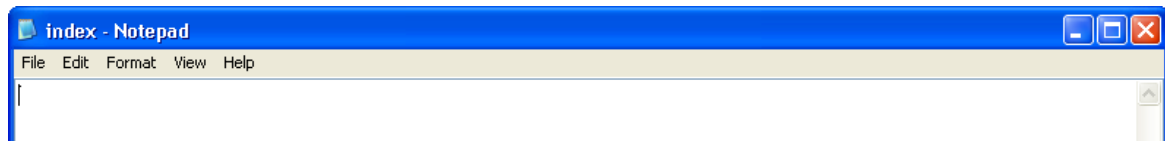
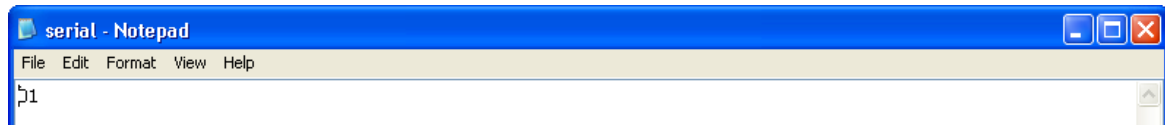
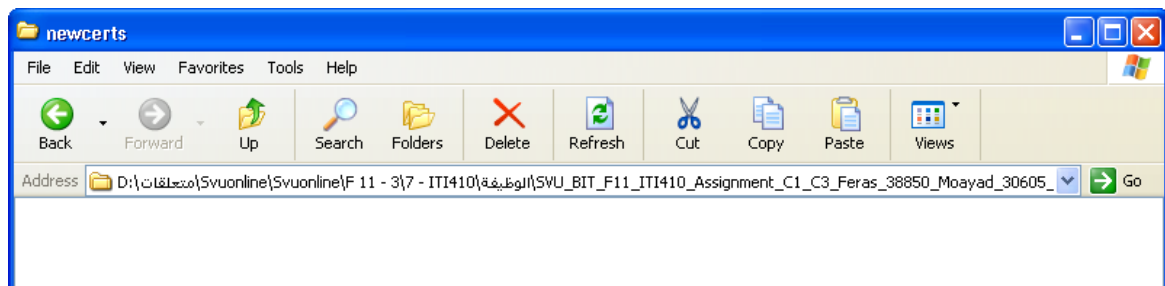
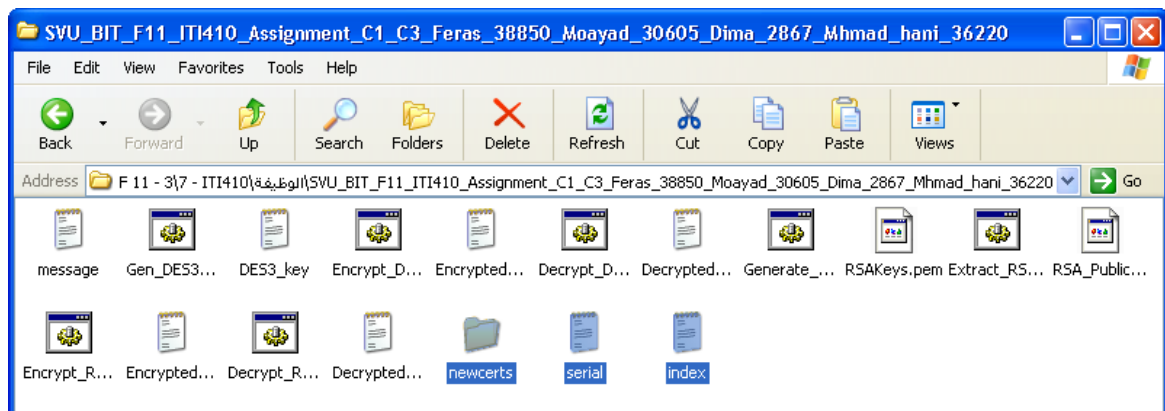
```
@echo off
call c:\openssl-win32\bin\openssl rsautl -in Encrypted_RSA_Public.txt
-out Decrypted_RSA_Private.txt -inkey RSAKeys.pem -decrypt
echo done
pause
```



code	Details
@echo off	command will either turn ON, or OFF, you put in a batch file from showing itself
call Path file	call file or program using full path
rsautl	command be used to sign, verify, encrypt and decrypt data using the RSA algorithm
-in file	options for getting file of available from path or name
-out file	option or your shell to redirect it to a file
-inkey file	the input key file, by default it should be an RSA private key.
-pubin	the input file is an RSA public key.
-decrypt	decrypt the input data using an RSA private key.
echo Done.	End echo
pause	Following the initial stop work when the user clicks on any button

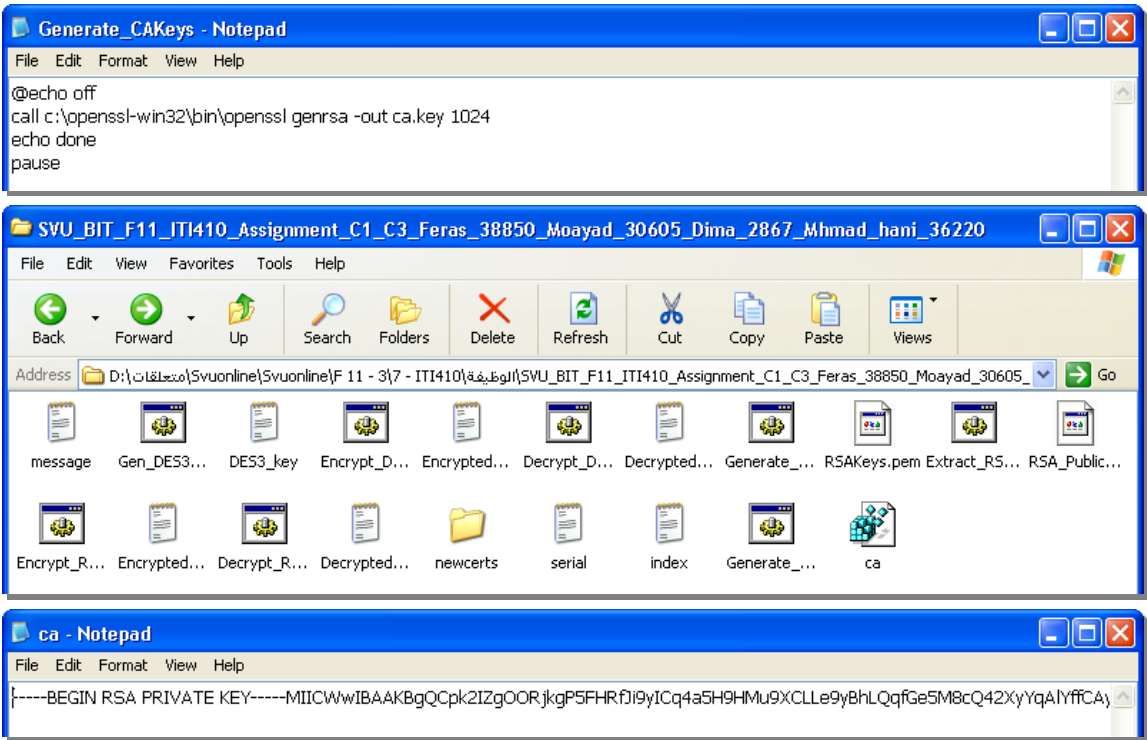
2-2. Certification

You will create a Certification Authority (CA) and two MIME clients. You need first to create a subfolder called "newcerts", a text file called "serial.txt" containing the text "01" and an empty text file called "index.txt".



9- Write a command line in a batch file called "Generate_CAKeys.bat" to generate the public/private key pair of the email Certification Authority (MIME CA) in PEM format. Run the command and save the result in a file called "ca.key".

```
@echo off
call c:\openssl-win32\bin\openssl genrsa -out ca.key 1024
echo done
pause
```

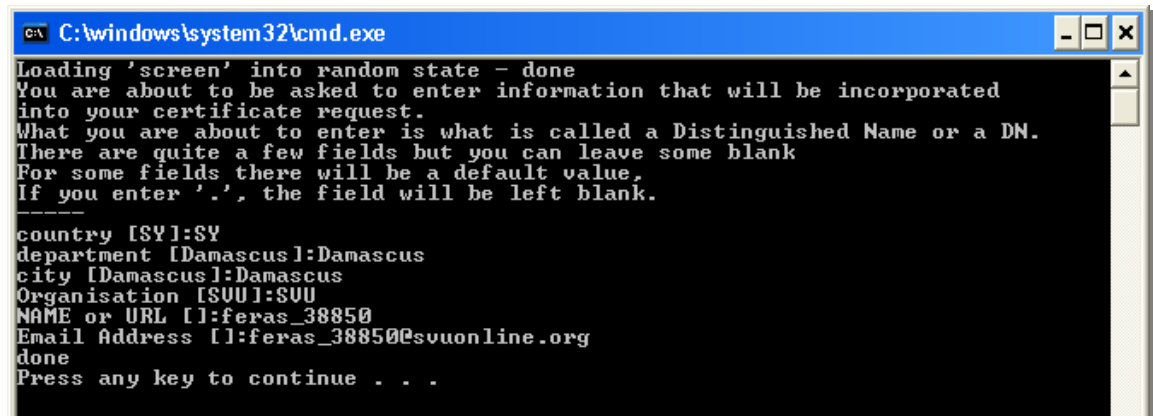
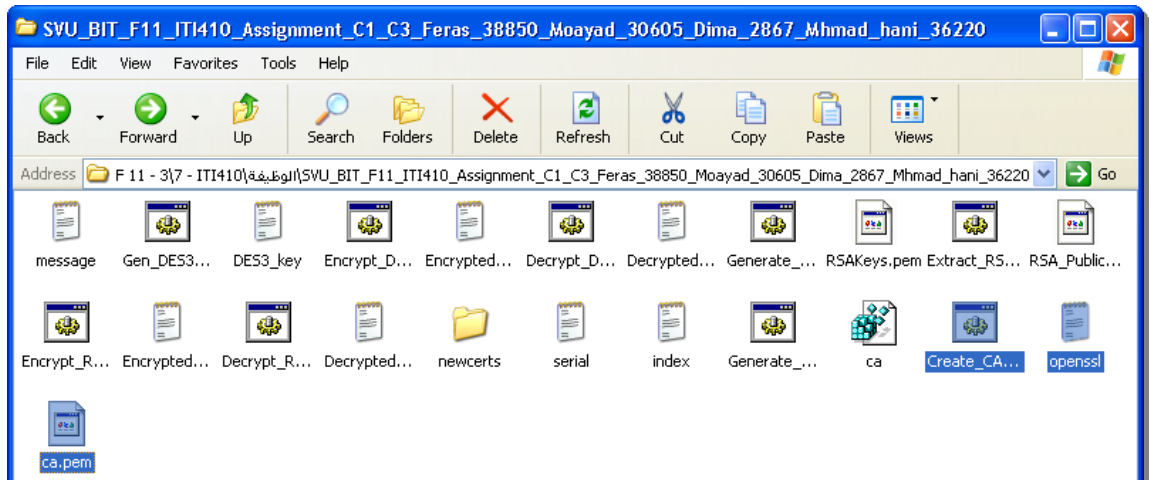
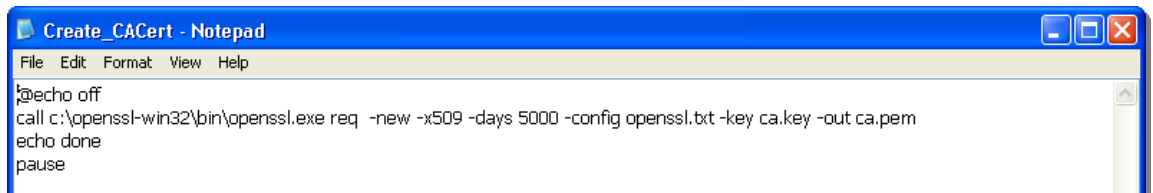


code	Details
@echo off	command will either turn ON, or OFF, you put in a batch file from showing itself
call Path file	call file or program using full path
genrsa	Standard commands , generate an RSA key
-out file 1024	File out size code is 1024 bits
echo Done.	End echo
pause	Following the initial stop work when the user clicks on any button

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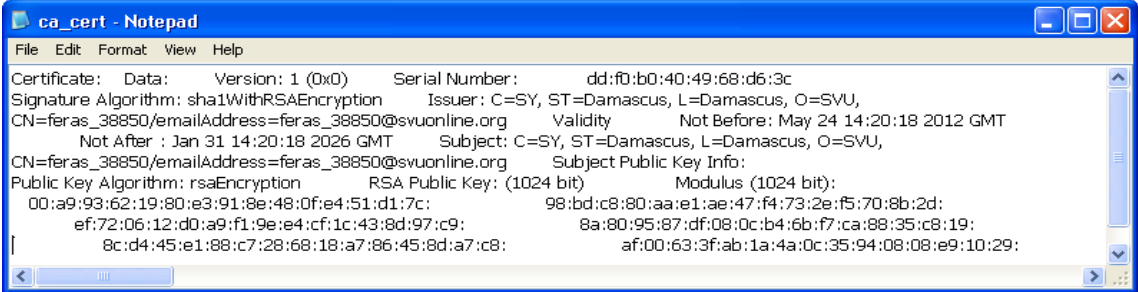
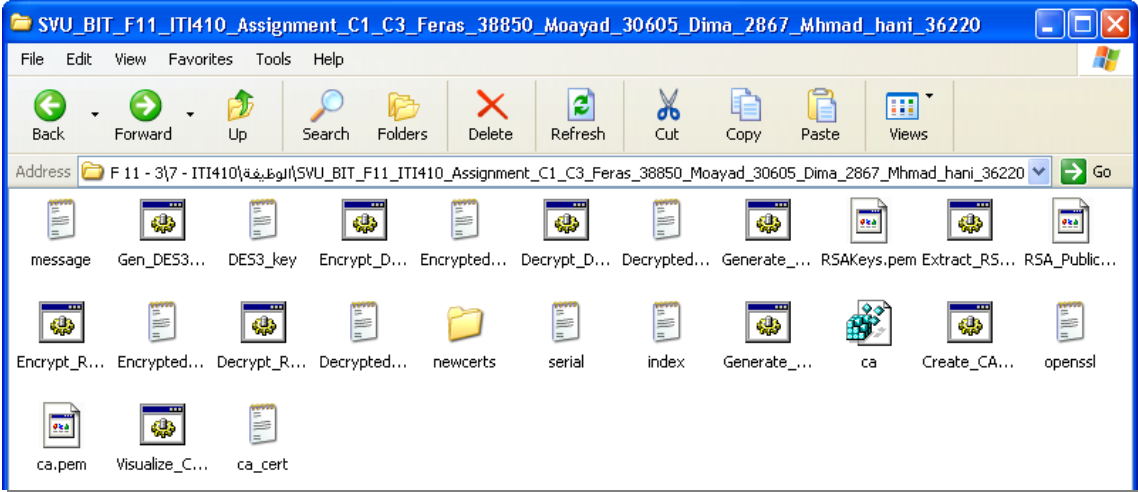
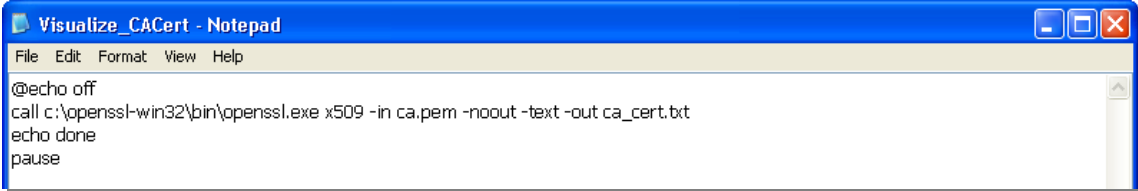
10- Write a command line in a batch file called "Create_CACert.bat" to create the auto-signed certificate of the CA in PEM format using the configuration file "openssl.txt" provided with the assignment. Run the command and save the result in a file called "ca.pem".

```
@echo off
call c:\openssl-win32\bin\openssl.exe req -new -x509 -days 5000 -
config openssl.txt -key ca.key -out ca.pem
echo done
pause
```



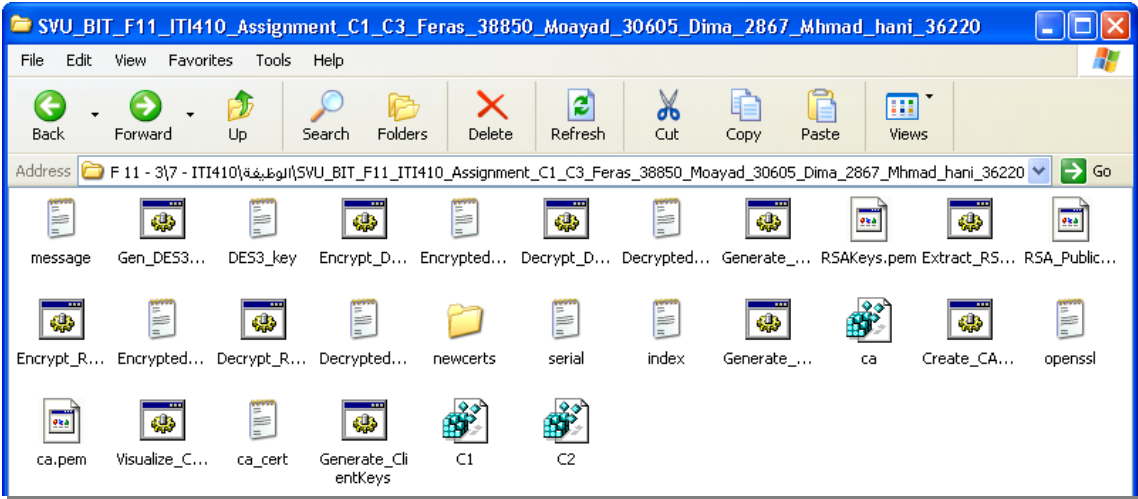
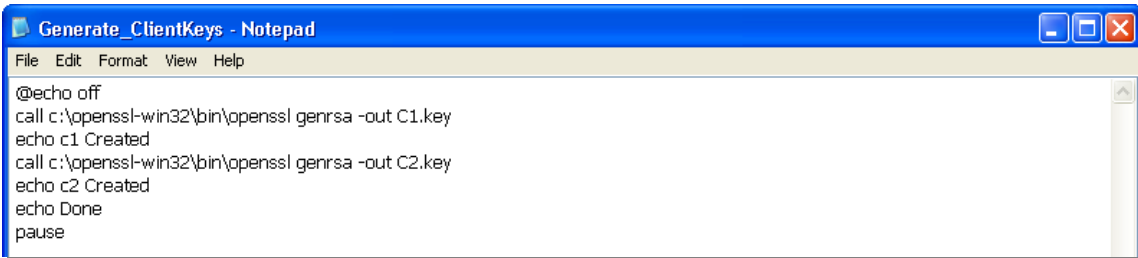
11- Write a command line in a batch file called "Visualize_CACert.bat" to create a readable version out of "ca.pem" to visualise the content of the auto-signed certificate of the CA. Run the command and save the result in a file called "ca_cert.txt".

```
@echo off
call c:\openssl-win32\bin\openssl.exe x509 -in ca.pem -noout -text -
out ca_cert.txt
echo done
pause
```



12- Write the needed command lines in a batch file called "Generate_ClientKeys.bat" to generate the public/private key pairs for two email clients (MIME clients) "C1" and "C2" in PEM format. Run the commands and save the results in two files called"C1.key" and "C2.key" respectively.

```
@echo off
call c:\openssl-win32\bin\openssl genrsa -out C1.key
echo c1 Created
call c:\openssl-win32\bin\openssl genrsa -out C2.key
echo c2 Created
echo Done
pause
```

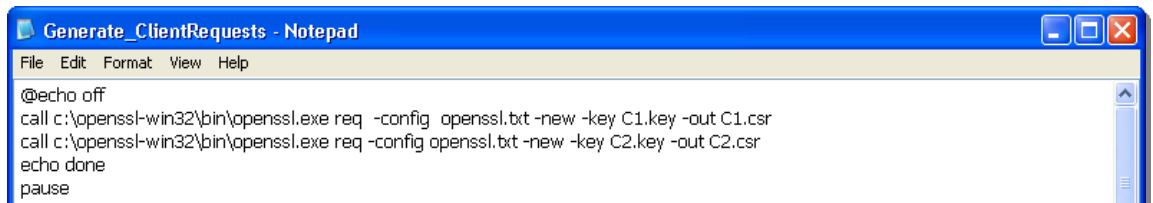


code	Details
genrsa	Standard commands , generate an RSA key
-out file	option or your shell to redirect it to a file
echo c1 Created	great file named is C1

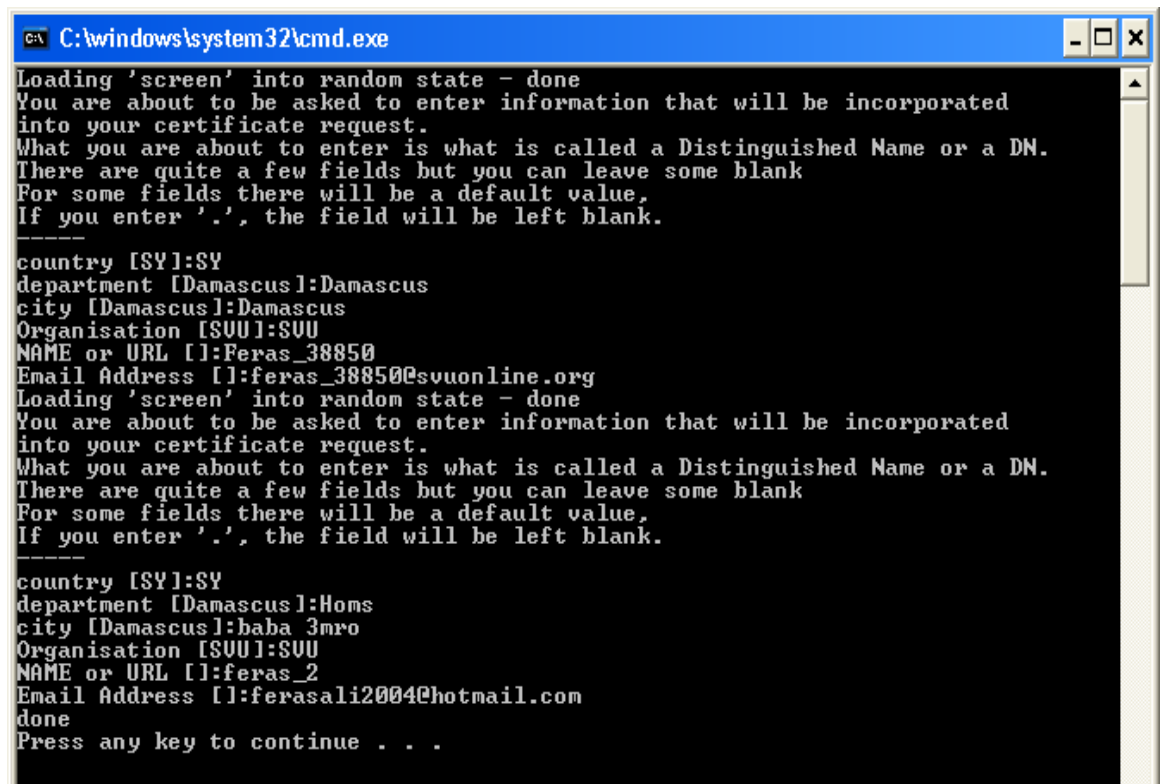
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- 13- Write the needed command lines in a batch file called "Generate_ClientRequests.bat" to generate a certificate request for each client using the configuration file "openssl.txt" provided with the assignment. Run the commands and save the results in two files called "C1.crs" and "C2.crs" respectively.

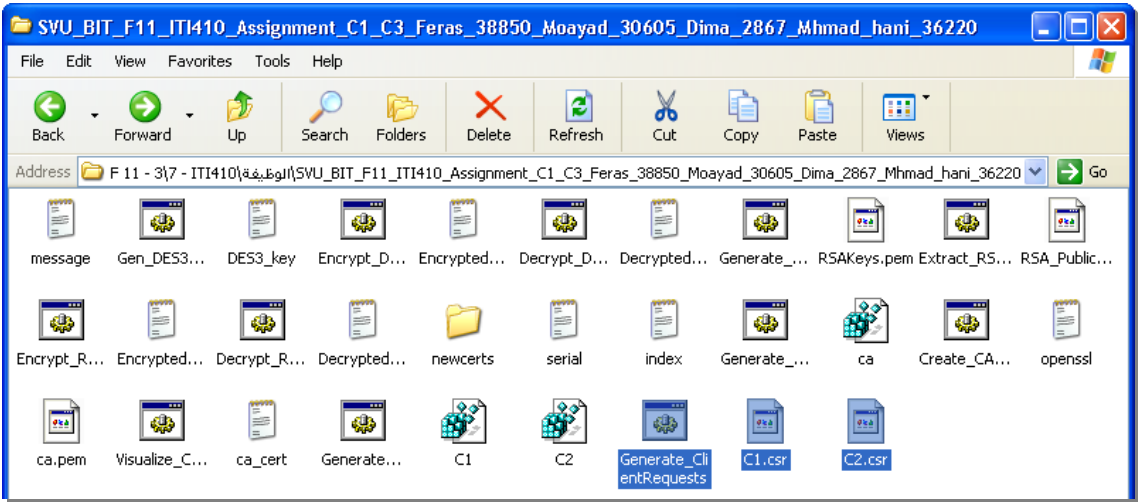
```
@echo off
call c:\openssl-win32\bin\openssl.exe req -config openssl.txt -new -
key C1.key -out C1.csr
call c:\openssl-win32\bin\openssl.exe req -config openssl.txt -new -
key C2.key -out C2.csr
echo done
pause
```



```
Generate_ClientRequests - Notepad
File Edit Format View Help
@echo off
call c:\openssl-win32\bin\openssl.exe req -config openssl.txt -new -key C1.key -out C1.csr
call c:\openssl-win32\bin\openssl.exe req -config openssl.txt -new -key C2.key -out C2.csr
echo done
pause
```



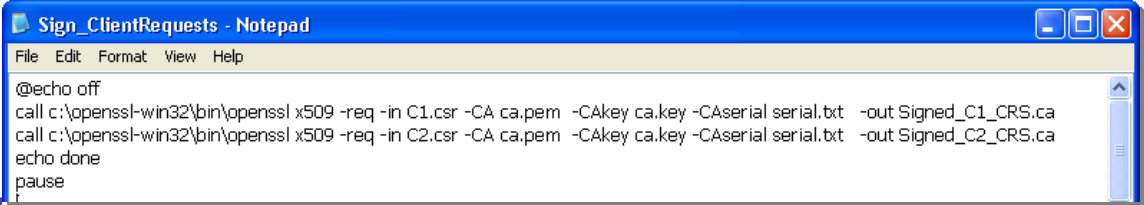
```
C:\windows\system32\cmd.exe
Loading 'screen' into random state - done
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
country [SY]:SY
department [Damascus]:Damascus
city [Damascus]:Damascus
Organisation [SVU]:SVU
NAME or URL []:Feras_38850
Email Address []:feras_38850@svuonline.org
Loading 'screen' into random state - done
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
country [SY]:SY
department [Damascus]:Homs
city [Damascus]:baba 3mro
Organisation [SVU]:SVU
NAME or URL []:feras_2
Email Address []:ferasali2004@hotmail.com
done
Press any key to continue . . .
```



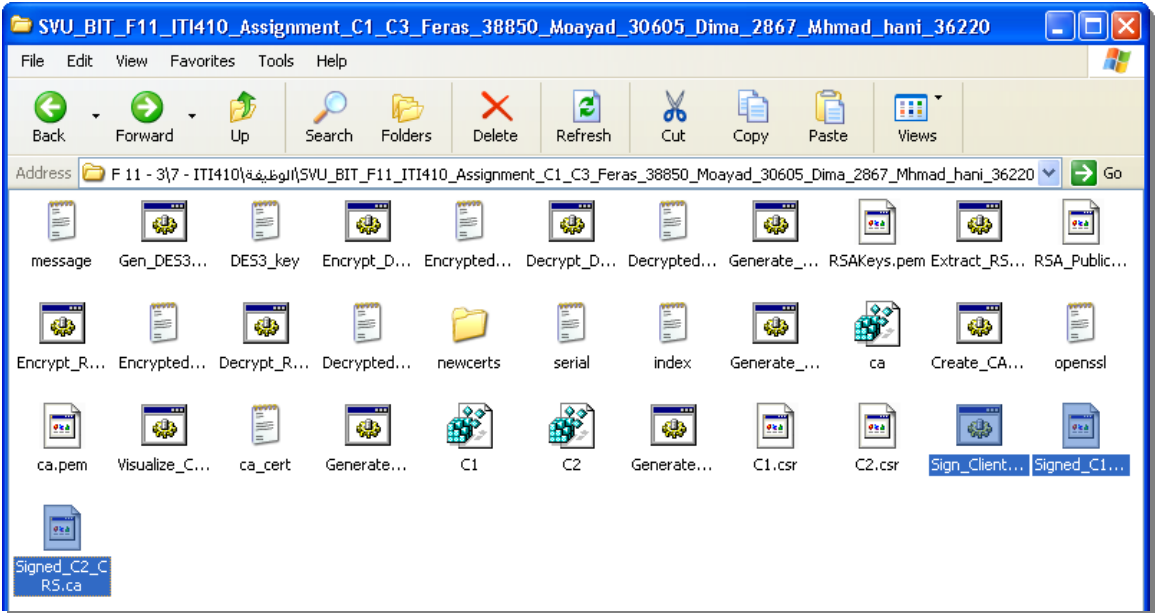
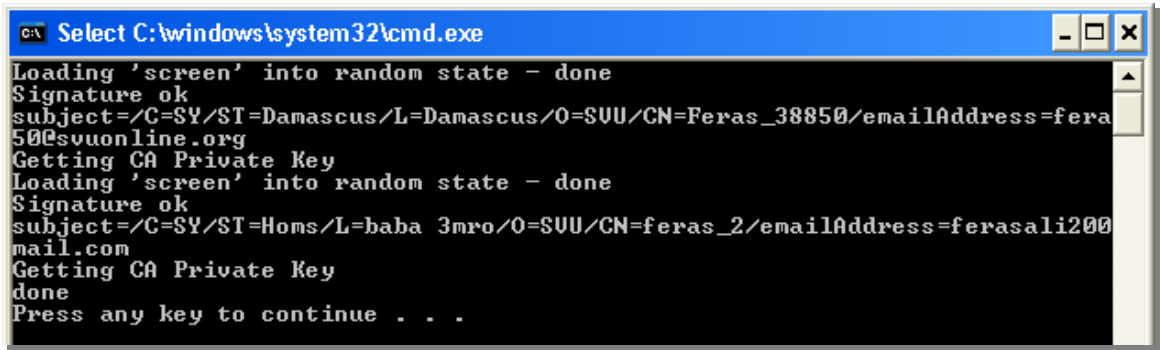
code	Details
req	PKCS#10 certificate request and certificate generating utility.
-config file	this allows an alternative configuration file to be specified, this overrides the compile time filename or any specified in environment variable.
-new	creates a new certificate request and a new private key
-key file	This specifies the file to read the private key from. It also accepts PKCS#8 format private keys for PEM format files.
-out file	option or your shell to redirect it to a file

14- The certificate requests should be sent to the CA to be signed, which should generate the client certificates. Taking the role of the CA, write the needed command lines in a batch file called "Sign_ClientRequests.bat" to sign the requests "C1.csr" and "C2.csr" of the two MIME clients, in order to generate their respective identity certificates. Run the commands and save the results in two files called "Signed_C1_CRS.ca" and "Signed_C2_CRS.ca" respectively.

```
@echo off
call c:\openssl-win32\bin\openssl x509 -req -in C1.csr -CA ca.pem -CAkey ca.key -CAserial serial.txt -out Signed_C1_CRS.ca
call c:\openssl-win32\bin\openssl x509 -req -in C2.csr -CA ca.pem -CAkey ca.key -CAserial serial.txt -out Signed_C2_CRS.ca
echo done
pause
```



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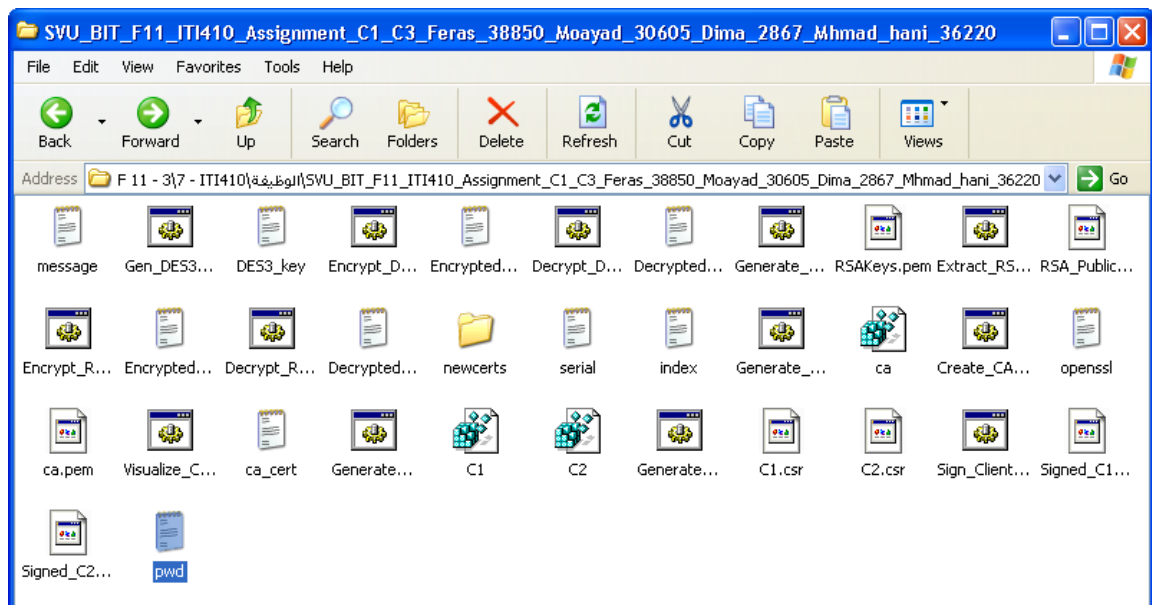


code	Details
x509	One of the types of certificates
-req	PKCS#10 certificate request and certificate generating utility.
-in file	options for getting file of available from path or name
-CA file	specifies the CA certificate to be used for signing. When this option is present x509 behaves like a ``mini CA''. The input file is signed by this CA using this option: that is its issuer name is set to the subject name of the CA and it is digitally signed using the CAs private key.
-CAkey file	sets the CA private key to sign a certificate with. If this option is not specified then it is assumed that the CA private key is present in the CA certificate file.
-CAserial file	sets the CA serial number file to use
-out file	option or your shell to redirect it to a file

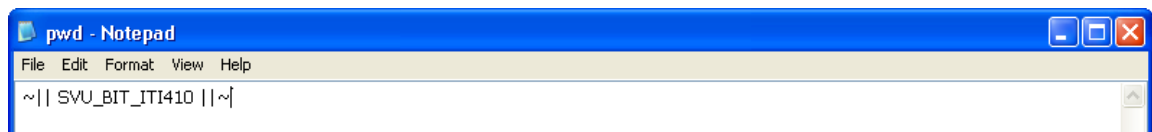
2-3. Sending Secure Messages

Client "C1" will send a message to client "C2" securely. Client "C1" will encrypt and sign his/her message and send it to client "C2".

15- Choose a password of 12 characters at least and store in a file called "pwd.txt".

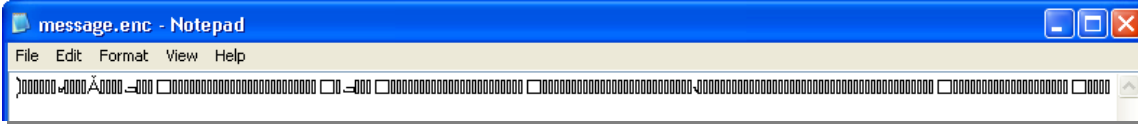
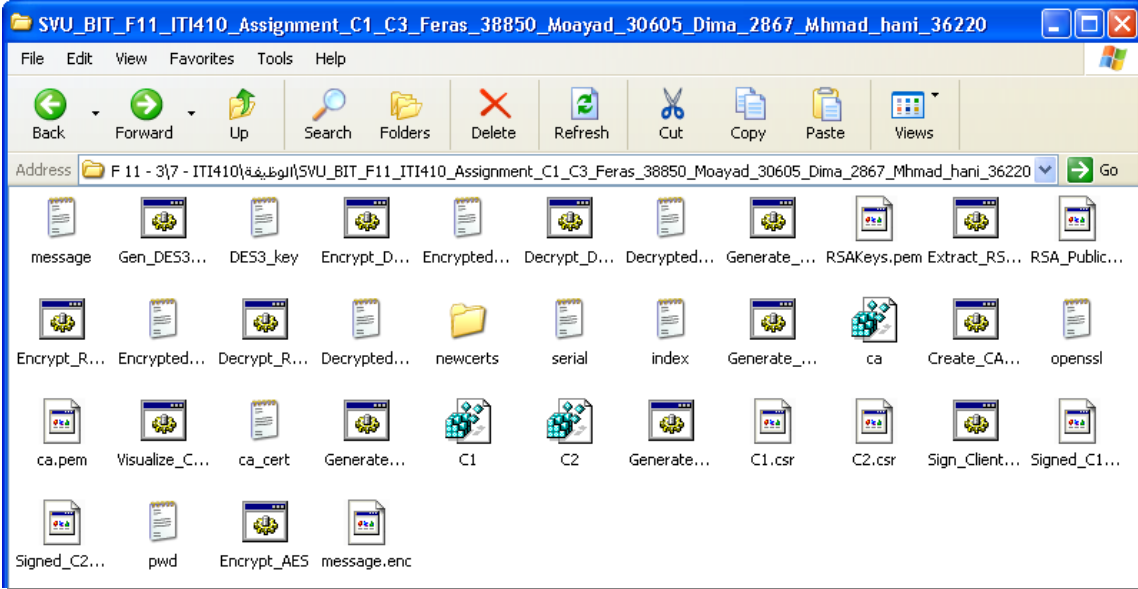
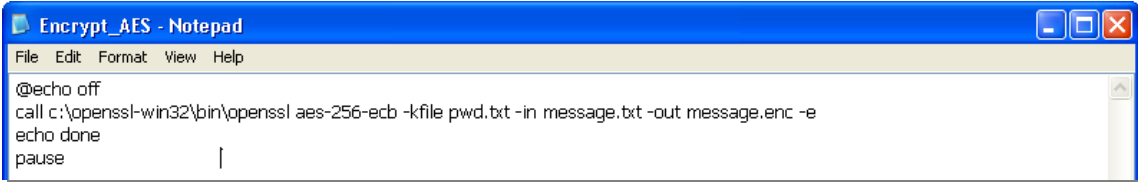


I choice the password : ~|| SVU_BIT_ITI410 ||~



16- Write a command line in a batch file called "Encrypt_AES.bat" to encrypt the message "message.txt" using the password stored in "pwd.txt" and the AES algorithm. Run the commands and save the results in a file called "message.enc".

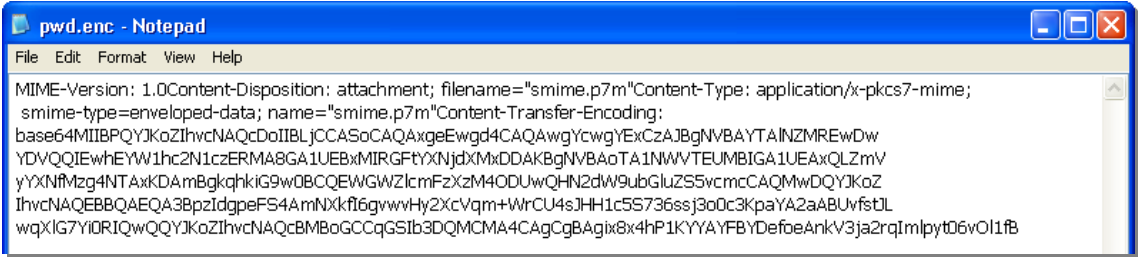
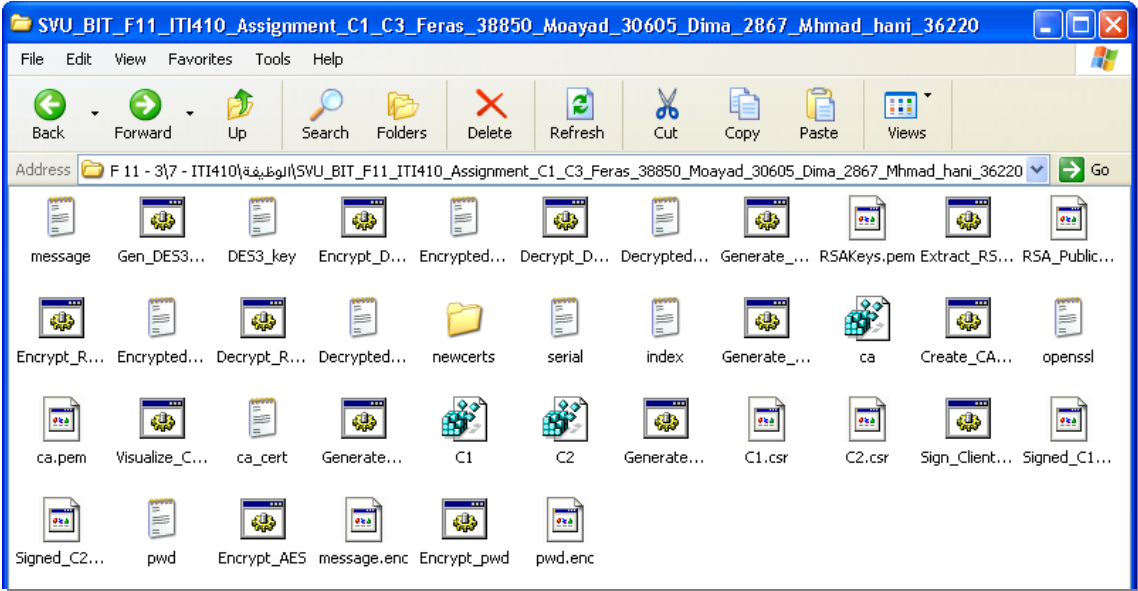
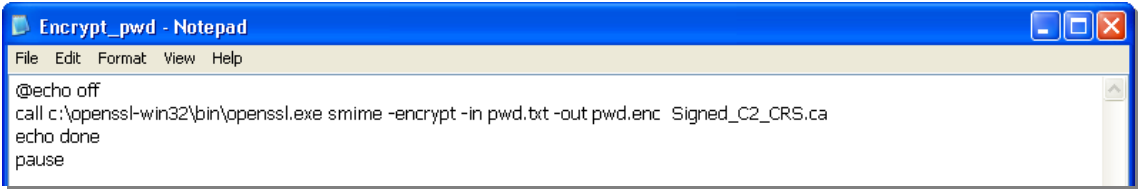
```
@echo off
call c:\openssl-win32\bin\openssl aes-256-ecb -kfile pwd.txt -in
message.txt -out message.enc -e
echo done
pause
```



code	Details
aes-256-ecb	Cipher commands (see the `enc` command for more details)
-kfile	
-in file	options for getting file of available from path or name
-out file	option or your shell to redirect it to a file

17- Write a command line in a batch file called "Encrypt_pwd.bat" to encrypt the password stored in "pwd.txt" using the public key of the client "C2" (his/her certificate). Use the command "smime" for this purpose. Run the command and save the result in a file called "pwd.enc".

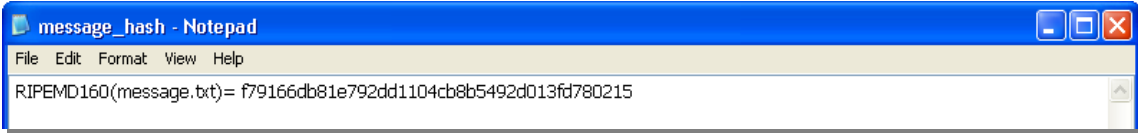
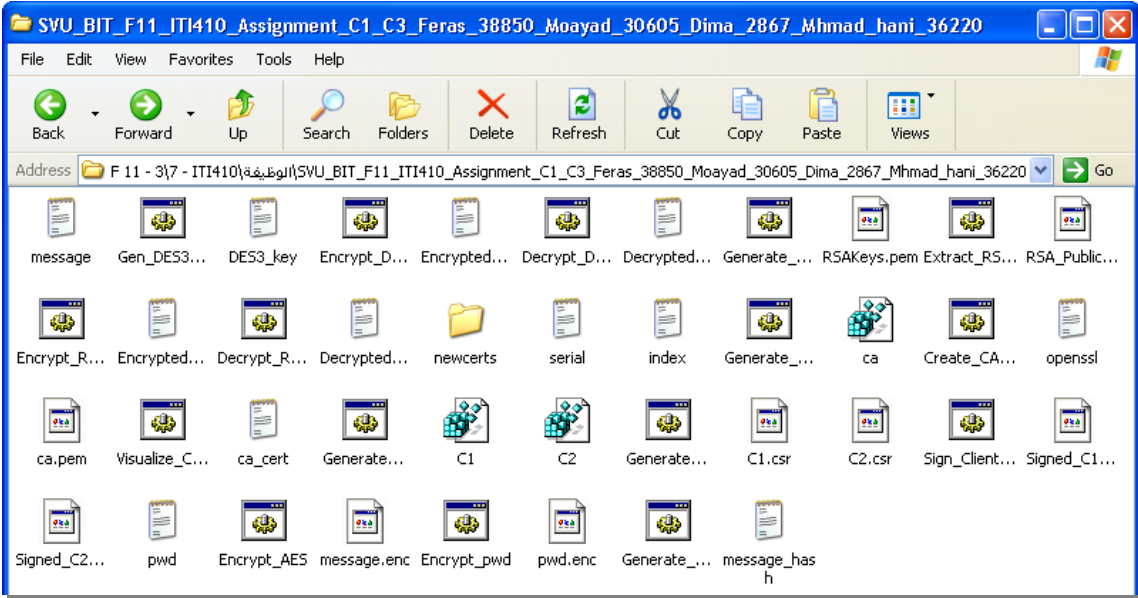
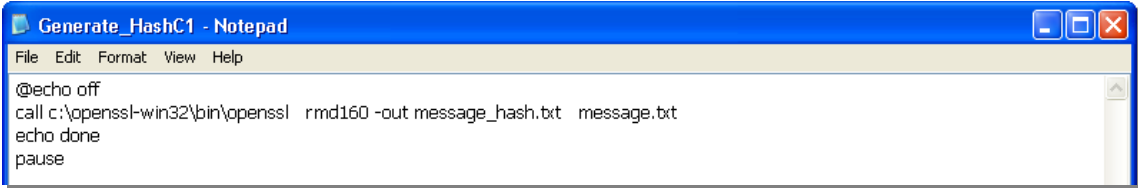
```
@echo off
call c:\openssl-win32\bin\openssl.exe smime -encrypt -in pwd.txt -out
pwd.enc Signed_C2_CRS.ca
echo done
pause
```



code	Details
smime	command handles S/MIME mail. It can encrypt, decrypt, sign and verify messages
-encrypt	encrypt the input data using an RSA public key
Signed_C2_CRS.ca	File : sign mail using the supplied certificate and private key. Input file is the message to be signed

18- Write a command line in a batch file called "Generate_HashC1.bat" to generate a hash value out of the message "message.txt" using the MD5 algorithm. Run the command and save the result in a file called "message_hash.txt".

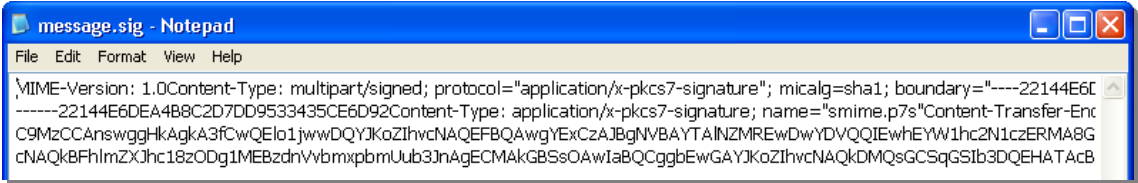
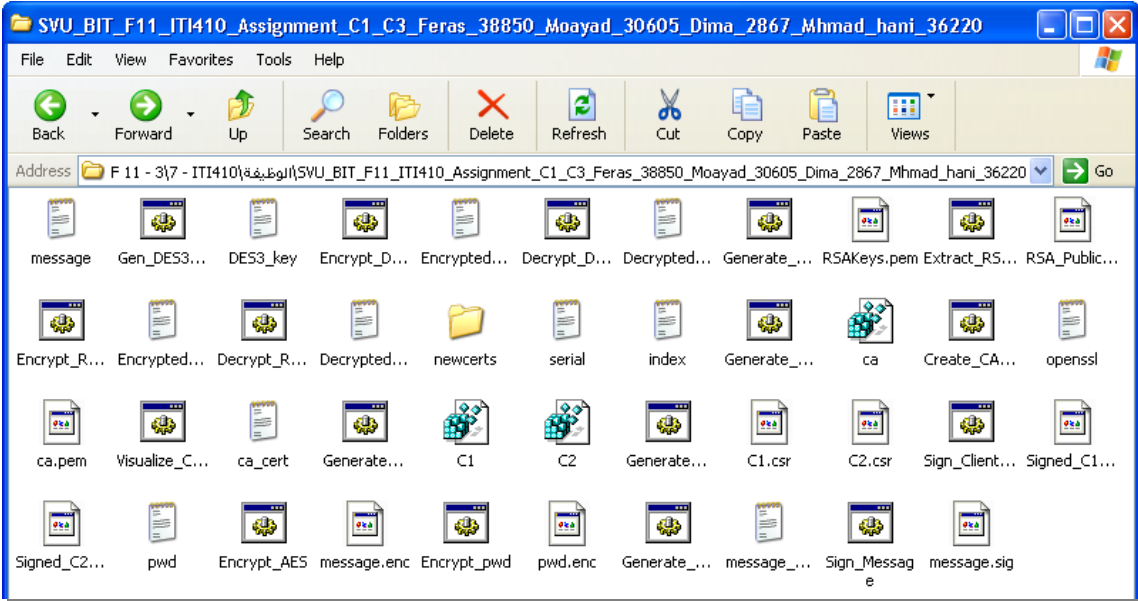
```
@echo off
call c:\openssl-win32\bin\openssl rmd160 -out message_hash.txt message.txt
echo done
pause
```



code	Details
rmd160	Message Digest commands ,13854.71k ,40271.23k ,87613.95k, 124333.06k
-out file	option or your shell to redirect it to a file
message.txt	Old file

19- Write a command line in a batch file called "Sign_Message.bat" to sign the hash value stored in "message_hash.txt" using your private key. Use the command "smime" for this purpose. Run the commands and save the results in a file called "message.sig".

```
@echo off
call c:\openssl-win32\bin\openssl smime -inkey C1.key -sign -in
message_hash.txt -out message.sig -signer Signed_C1_CRS.ca -certfile
ca.pem
echo done
pause
```



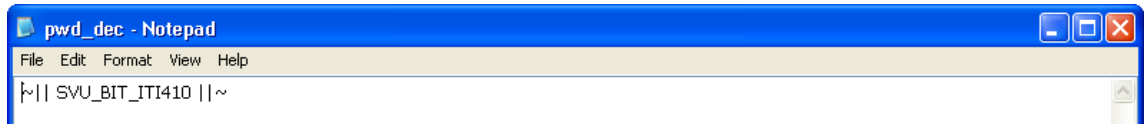
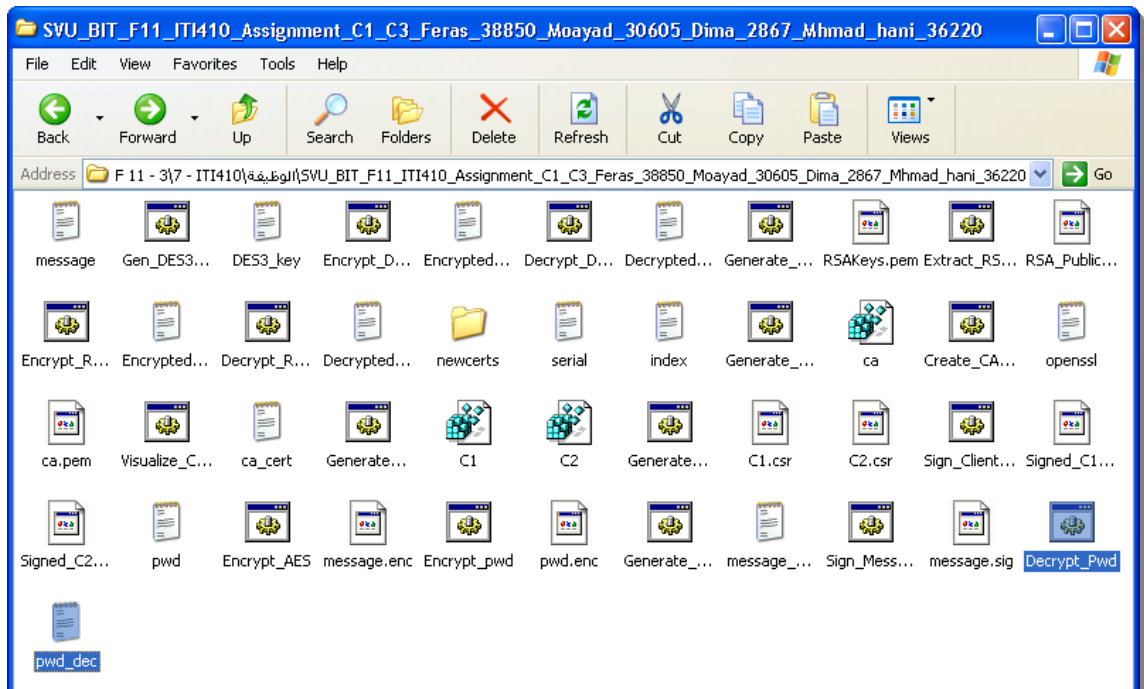
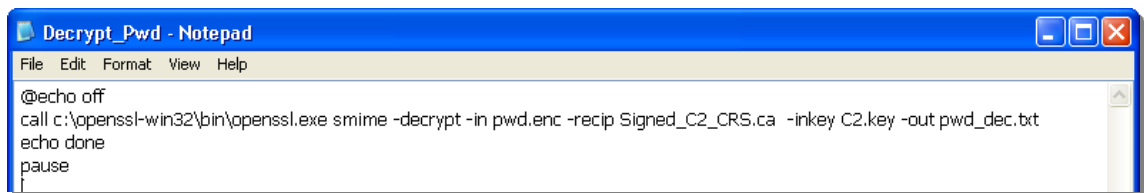
code	Details
smime	standard for sending and receiving secure MIME data
-sign	sign digest using private key in file
-signer file	Name file the signer's public key
-certfile ca.pem	Old file

2-4. Receiving Secure Messages

Client "C2" will receive the encrypted and signed message sent by client "C1". It will receive the encrypted message text "message.enc", the encrypted password "pwd.enc" and the message signature "message.sig". Client "C2" will decrypt and verify the received message.

20- Write a command line in a batch file called "Decrypt_Pwd.bat" to decrypt the encrypted password "pwd.enc" using the private key of client "C2". Use the command "smime" for this purpose. Run the command and save the result in a file called "pwd_dec.txt".

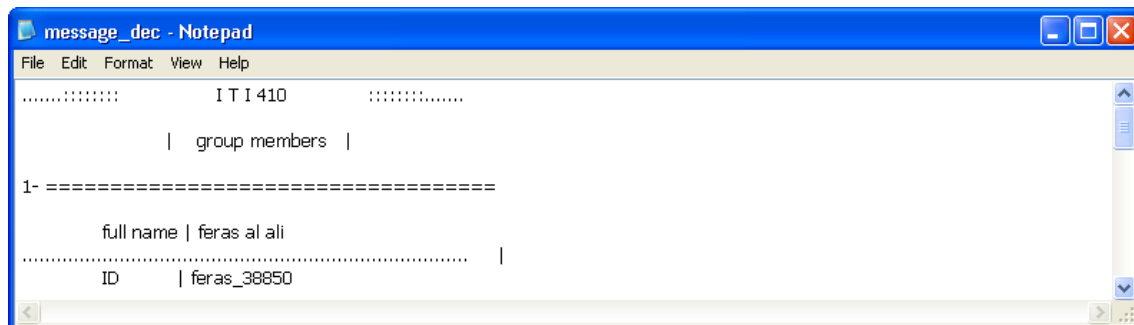
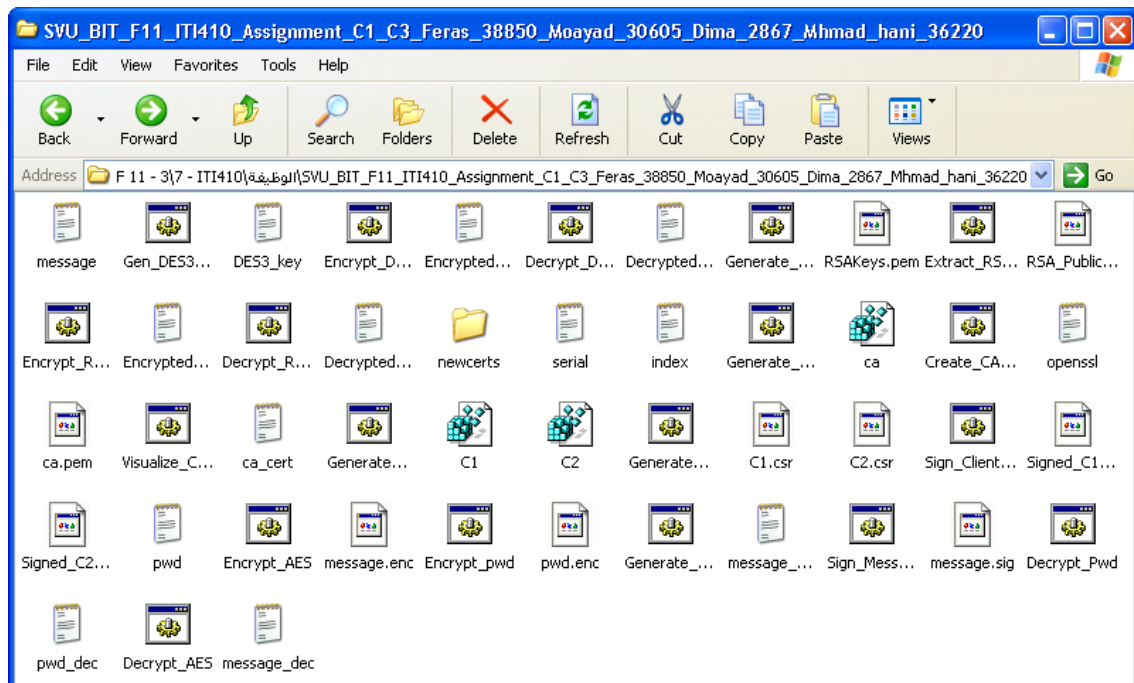
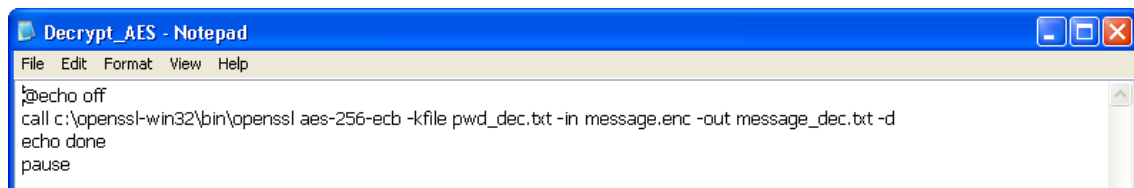
```
@echo off
call c:\openssl-win32\bin\openssl.exe smime -decrypt -in pwd.enc -
recip Signed_C2_CRS.ca -inkey C2.key -out pwd_dec.txt
echo done
pause
```



.....

21- Write a command line in a batch file called "Decrypt_AES.bat" to decrypt the encrypted message "message.enc" using the password stored in the file "pwd_dec.txt" and the AES algorithm. Run the commands and save the result in a file called "message_dec.txt".

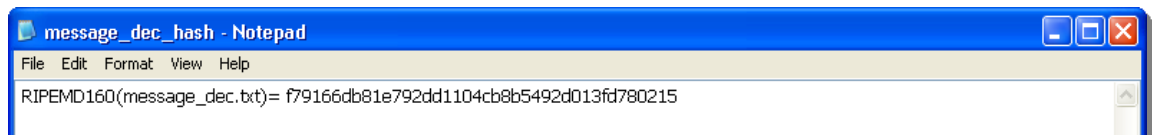
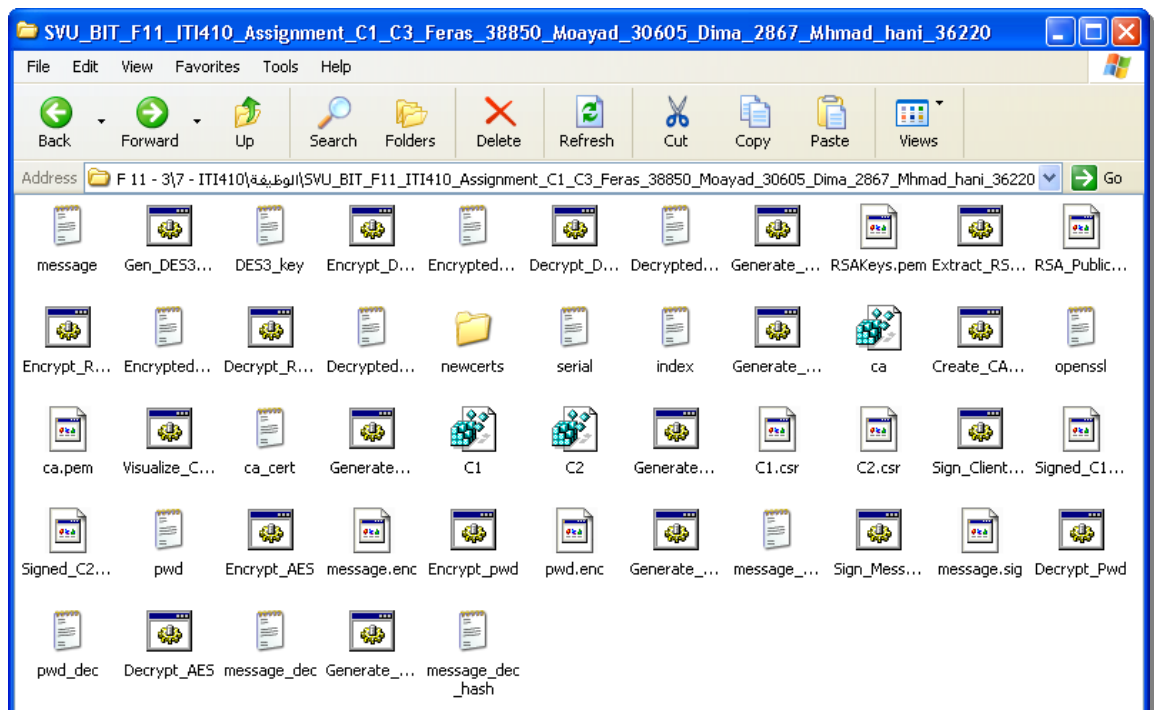
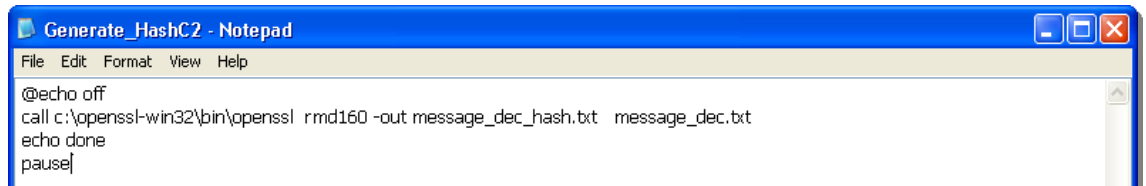
```
@echo off
call c:\openssl-win32\bin\openssl aes-256-ecb -kfile pwd_dec.txt -in
message.enc -out message_dec.txt -d
echo done
pause
```



.....

22- Write a command line in a batch file called "Generate_HashC2.bat" to generate a hash value out of the message "message_dec.txt" using the MD5 algorithm. Run the commands and save the result in a file called "message_dec_hash.txt".

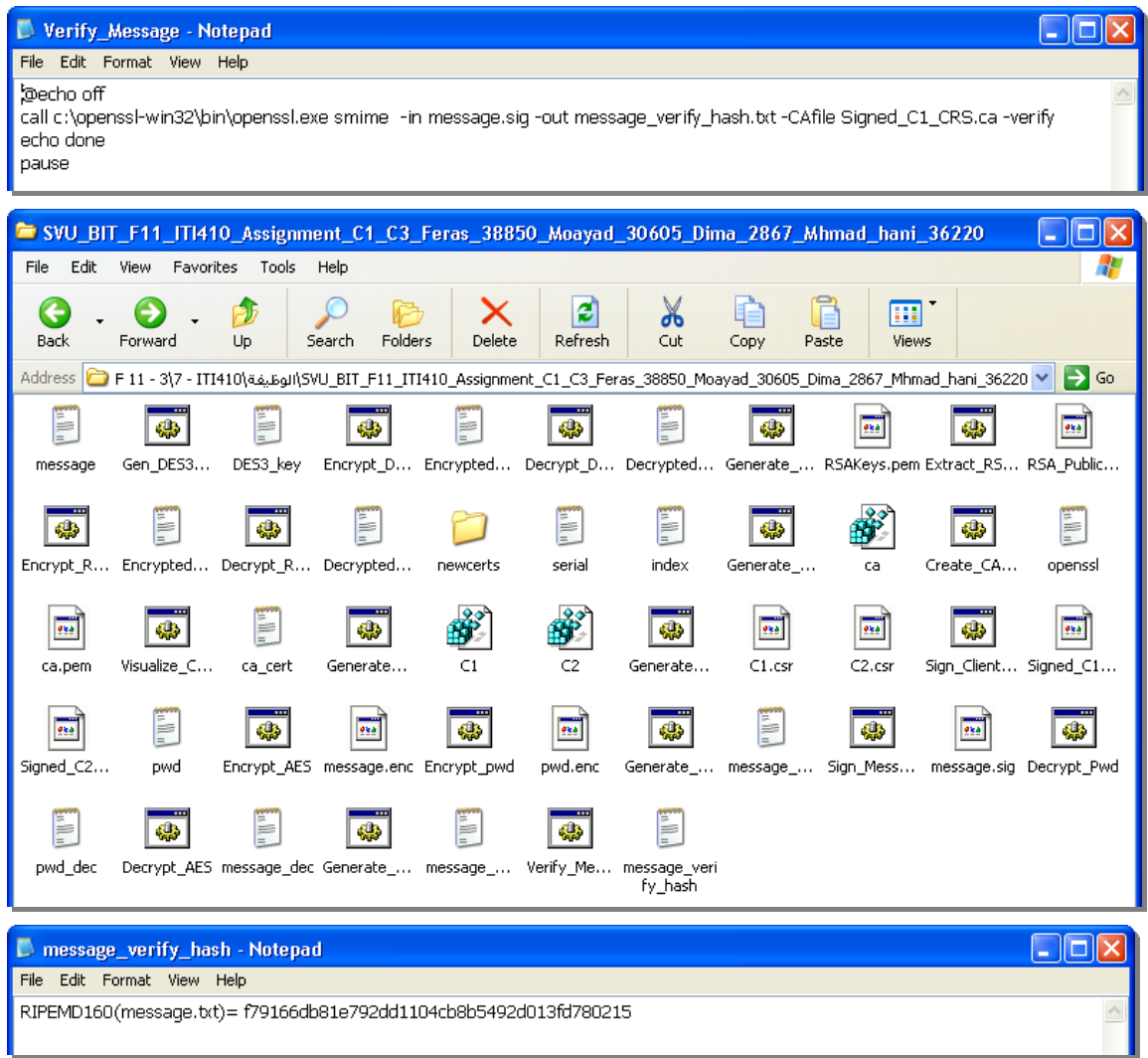
```
@echo off
call c:\openssl-win32\bin\openssl rmd160 -out message_dec_hash.txt
message_dec.txt
echo done
pause
```



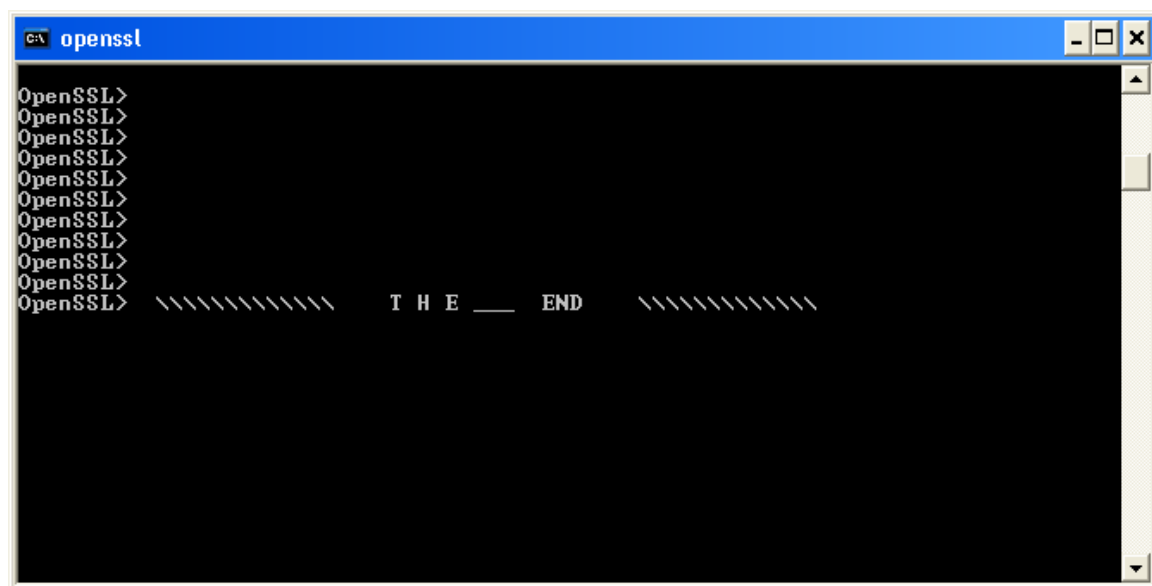
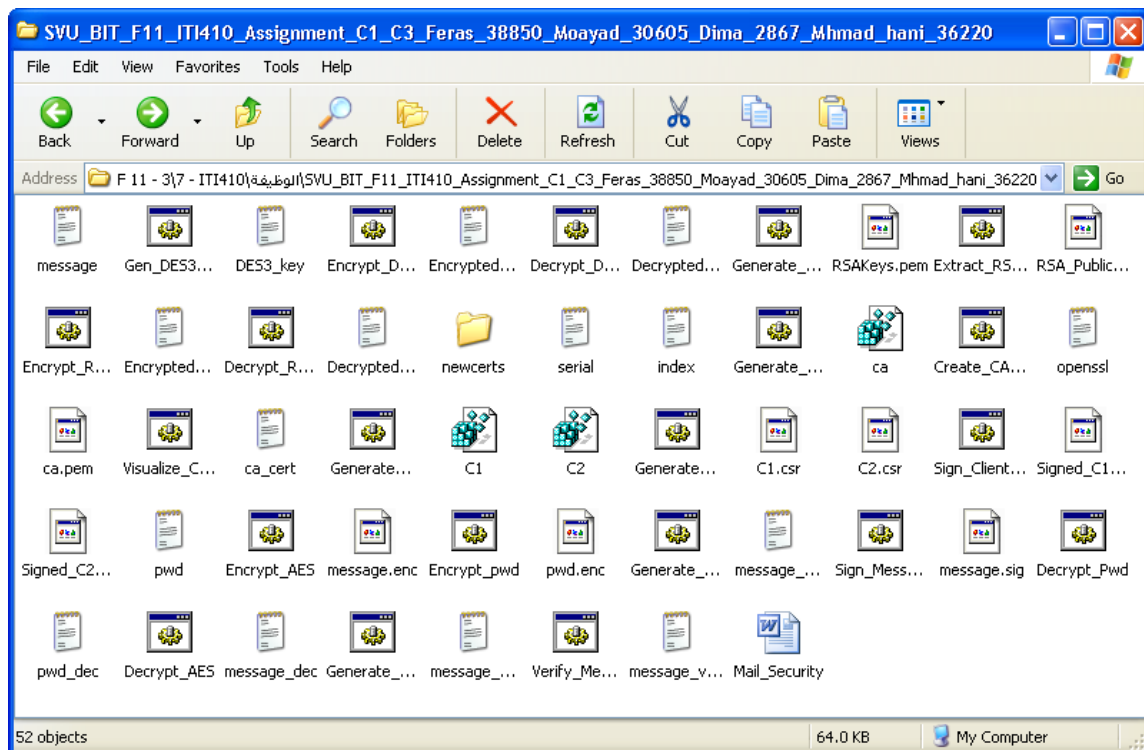
.....

23- Write a command line in a batch file called "Verify_Message.bat" to verify the signature stored in the file "message.sig" using the public key of the client "C1" (his/her certificate). Use the command "smime" for this purpose. Run the commands and save the result in a file called "message_verify_hash.txt".

```
@echo off
call c:\openssl-win32\bin\openssl.exe smime -in message.sig -out
message_verify_hash.txt -CAfile Signed_C1_CRS.ca -verify
echo done
pause
```



24- Explain in a Word document called "Mail_Security.doc" the security concepts (Confidentiality, Integrity and Availability) that this secure mail exchange model achieves. Also identify if any vulnerabilities could exist.



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